

Sparse-Autoencoder Dissection of Emotion Features in Frozen Speech Encoders

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Abstract

Linear probes show emotion is decodable in the middle-to-late layers of frozen speech transformers, but not which internal features carry it, nor whether the signal is acoustic or lexical. We train TopK Sparse Autoencoders (SAEs) on *per-frame* activations of six frozen large encoders and dissect their emotion features on the lexicon-controlled CREMA-D corpus (91 actors $\times$ 12 fixed sentences $\times$ 6 emotions). A confound-controlled selection—a feature’s firing must be more informative about emotion than about the spoken sentence or the speaker—isolates genuine emotion features from phonetic and speaker-driven ones. We find that (i) emotion *disentangles* into clean monosemantic features earlier (L2–L7) than where it is best linearly *decoded* (L10); (ii) all six encoders share an acoustic emotion-feature basis that generalizes across unseen speakers; and (iii) the recovered emotion signal is acoustic, not lexical. Decodability and disentanglement are thus distinct, and emotion here is carried by speaker-invariant acoustic features.

Keywords

speech emotion recognition, sparse autoencoders, mechanistic interpretability, probing, acoustic features, disentanglement

1 Introduction

Self-supervised speech encoders dominate speech emotion recognition (SER), yet were trained on ASR-style objectives that encode substantial linguistic structure. Layer-wise probing answers *where* emotion is readable, but cannot say *which* features encode it, whether the depth pattern holds across encoders, speakers and languages, or whether the signal a probe exploits is acoustic (how something is said) or lexical (what is said). Sparse autoencoders (SAEs) decompose dense activations into an over-complete dictionary of sparse, near-monosemantic features [2, 5], turning these questions from decodability into mechanism. Because SAEs trained on generic speech tend to rediscover *phonemes* rather than affect [1], we anchor the study on CREMA-D [3], whose 12 fixed carrier sentences hold text constant across emotions: any emotion signal recovered there is necessarily acoustic, giving a built-in positive control. We contribute (1) a confound-controlled emotion-feature definition for speech SAEs, and a three-part answer showing that (2) disentanglement and decodability peak at different depths, and (3) the emotion features are acoustic and speaker-invariant.

2 Method

Per-frame activations. The cached embeddings are mean-pooled per utterance—too few to fit an $8\times$ over-complete SAE—so we re-extract *un-pooled* per-frame hidden states from the six frozen large encoders (HuBERT [6], WavLM, wav2vec2, w2v-BERT, MERT, Whisper [8]): 25 states per frame (CNN L0 + 24 transformer layers), Whisper padding masked. CREMA-D layer 13 alone gives 940,273

frames. We train a TopK SAE ($d_{in}=1024$, $d_{sae}=8192$, $k=32$) with unit-norm decoder, input standardisation, and an AuxK loss reviving dead features; it reaches FVU = 0.070 (93% variance explained) with 4/8192 dead features.

Confound-controlled emotion features. We score each feature by the mutual information (MI) between its firing and the emotion label, thresholded at 5σ over a label-permutation null. Significance plus monosemanticity (≥ 0.5 mass on one emotion) flags 394 features—but $\sim 66\%$ carry *more* information about the spoken sentence (phonetic content) or the speaker than about emotion. Requiring $MI(\text{emotion}) > MI(\text{sentence})$ and $> MI(\text{actor})$ removes them, leaving 61 genuine emotion features (median monosemanticity 0.59), dominated by anger and sadness; fear and neutral fall to zero, consistent with acoustic encoding of high-arousal emotions. For each, we compare the cross-validated ROC-AUC of a logistic classifier predicting its firing from eGeMAPS descriptors [4] (acoustic) vs. carrier-sentence identity (lexical); R^2 is unusable as the features are $\sim 96\%$ zero.

3 Results

3.1 Depth: decode $\neq$ disentangle (Q i)

A 25-layer sweep separates two curves (Fig. 1, top-left; Table 1). Probe accuracy peaks at L10 and stays high into the deepest layers, but the *count* of monosemantic emotion features peaks earlier at L7 (and monosemanticity at L2). The deep layers L17–24 retain decodability (~ 0.72) yet collapse to 24–47 emotion features: emotion stays linearly readable while becoming distributed rather than packed into clean directions. *It organises into interpretable features before the depth at which a probe reads it best.*

3.2 Across paradigms and speakers (Q ii)

All six encoders carry emotion features at their best layer (Fig. 1, bottom-left), each anger/sadness-dominated. Matching features by their per-utterance firing correlation, the Whisper $\rightarrow$ SSL overlap (0.40) equals the SSL $\rightarrow$ SSL overlap (0.39): on fixed-lexicon CREMA-D Whisper is *not* distinct, so its known divergence concerns the lexical shortcut, not acoustic emotion encoding. Under speaker-disjoint cross-validation across the 91 actors emotion accuracy is unchanged (leave-speaker-out drop ≈ 0 ; $\sim 95\%$ actor coverage); confound control already removed the speaker-entangled features behind the published MESD leave-one-speaker-out collapse (~ 0.30).

3.3 Acoustic, not lexical (Q iii)

The positive control (fixed lexicon $\Rightarrow$ acoustic) labels 60/61 emotion features acoustic (eGeMAPS firing-AUC 0.83 vs. sentence 0.66). Causally (Fig. 1, bottom-right), emotion decodes from the 61 acoustic features at 0.552, vs. 0.431 from lexical and 0.423 from random features (6-class; full reconstruction 0.73). Since the signal is distributed, sufficiency rather than ablation is the informative test: the recovered emotion signal is acoustic.

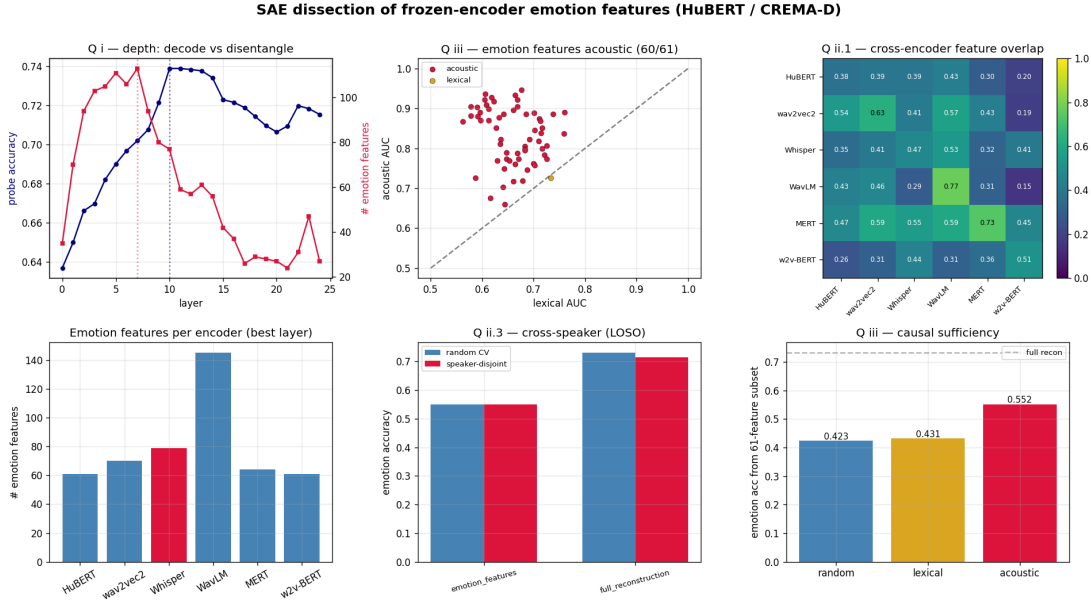


Figure 1: SAE dissection of frozen-encoder emotion features (HuBERT / CREMA-D). Top, left to right: depth decode-vs-disentangle (probe accuracy vs. # monosemantic emotion features per layer); emotion features acoustic vs. lexical (eGeMAPS vs. sentence firing-AUC, points above the diagonal are acoustic); cross-encoder feature-overlap matrix. Bottom: emotion features per encoder at its best layer; cross-speaker (leave-speaker-out) generalization vs. random CV; causal sufficiency of acoustic vs. lexical vs. random feature subsets.

4 Discussion and Limitations

Decodability and disentanglement are distinct axes: a probe reads emotion best where the representation is linearly separable (L10 and the deep layers), but the encoder packs emotion into clean monosemantic features earlier (L2–L7), sharpening the prior “middle-to-late” decodability result into a claim about where emotion is organised. On lexicon-controlled data all six encoders converge on a shared acoustic basis whose features survive speaker-disjoint evaluation; the Whisper-vs-SSL divergence concerns the lexical shortcut [7], which CREMA-D cannot pose and the deferred EMIS test targets. Three experiments await cluster data, each with a runnable

stub: the IEMOCAP real-transcript ablation, the SpeechCraft cross-lingual arm, and the EMIS audio-vs-text separation. All artifacts—activations, SAEs, per-step CSVs/plots/metrics, and this paper—are released for reproduction.

References

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Table 1: Key quantities (HuBERT / CREMA-D, layer 13 unless noted).

Quantity	Value
SAE reconstruction FVU (variance expl.)	0.070 (93%)
SAE size / sparsity / dead features	8192 / $k=32$ / 4
Emotion feats: sig → mono → confound	6407 → 394 → 61
Decode peak / disentangle peak / mono peak	L10 / L7 / L2
Cross-encoder overlap: Whisper→SSL / SSL→SSL	0.40 / 0.39
Acoustic features (positive control)	60/61
mean firing-AUC acoustic / lexical	0.83 / 0.66
Sufficiency: acoustic / lexical / random	0.552/0.431/0.423
Cross-speaker LOSO drop / actor coverage	≈ 0 / 0.95